Digital Forensics in Cloud Computing: Challenges, Techniques, and Legal Implications

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Abstract

Cloud computing now dominates daily activities for everyone from personal use to business operations. The use of cloud services through desktop, tablet, or smartphones provides essential data management and security functions. As our dependence on these services grows stronger, investigators face increasing difficulties especially when handling digital evidence.

Digital forensics analysts are facing server challenges during investigating some critical data that is stored in multi trenet systems. Evidence might be difficult to locate and collect because it might be stored in multiple servers, protected by complicated encryption and regulated by different legal systems.

The research investigates these problems by studying data dispersal challenges and encryption and jurisdiction complications and evaluates contemporary forensic methods and tools operating in cloud environments. The research project investigates legal challenges involved with digital evidence discovery from cloud sources and follows up with analysis of evidence handling during investigation through to presentation.

The purpose of this research project is to develop a conceptual framework that can speed up the forensic process and reduce investigation time while meeting legal requirements by reviewing existing literature and industry best practices. The study aims at identifying best practice recommendations for cloud computing digital forensics with the aim of contributing to technical innovation and policy development in this rapidly evolving field.

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Chapter I — Introduction

1.1 Introduction and Overview

Cloud computing has changed the way that data is managed completely, and it has become a critical factor in both personal and enterprise management. It has been estimated that the expenditure by end users on public cloud services across the world will be $723.4 billion in the year 2025, thus growing at a compound annual growth rate (CAGR) of 21.5% from the year 2024 and the outlay of $595.7 billion **(Gartner, 2024)**. This transformation is enabled by AWS, Azure and Google Cloud, which allow sharing data between different devices **(Daniel et al., 2024).** However, this will bring significant challenges to the digital forensics field, impacting the preservation, identification, analysis, and presentation of digital forensics for legal purposes. The architectures of cloud computing are dynamic, distributed, and vastly different from the static, centralized systems for which traditional forensic tools and approaches were designed, making the evidence collection and analysis process complicated **(Montasari and Hill, 2019)**.

Traditional forensics is much easier than cloud computing which introduces complexities like multi-tenancy and data dispersion across global data centers **(Cinar and Bharadiya, 2023)**.
For instance, a forensics investigation can involve different servers that are located in the U.S., the EU, and Asia. These servers are governed by different legal legislations like the GDPR or the CLOUD Act **(REGULATION (EU), 2016).** The 2024 Thales Data Threat Report shows that 93% of organizations had a cloud-related data breach, and the risks are rising in these environments **(Thales, 2024)**. Moreover, the global average cost of a data breach in 2024 was $4.88 million **(IBM, 2024)**, which shows that such incidents are costly for the businesses. Although encryption enhances security, it obstructs the access for digital forensics, and it also requires the cooperation from cloud service providers (CSPs) **(Montasari and Hill, 2019)**. These issues are not fully addressed by legal frameworks. The tension between these issues is evident in the Microsoft Ireland case, where US authorities attempted to obtain data from Ireland, raising data sovereignty and privacy issues that have not been resolved to this day, despite subsequent legislation **(Microsoft, no date)**. This dissertation aims to identify innovations that can improve cloud forensic investigations by exploring these technical and legal challenges.

1.2 Aims and Objectives

This dissertation aims to explore the details of digital evidence collection in cloud computing environments, evaluate current forensic tools and methods, and examine the legal and regulatory issues. This dissertation aims to answer the following questions:

- What specific challenges limit effective digital forensic investigations in cloud environments?
- How effective are the current forensic techniques and tools in addressing these cloud-specific challenges?
- To what extent do current legal and regulatory frameworks support or obstruct the process of collecting, preserving, and presenting digital evidence from cloud-based systems?
- What methodologies are currently employed in cloud-based digital forensic investigations and what gaps exist in these practices?
- What solutions and best practices can be proposed to enhance the efficiency, reliability, and legal compliance of digital forensic investigations in cloud computing?

In this literature review, I will examine current technical challenges, forensic tools and methodologies, and legal issues that are related to cloud-based digital forensics. Following this analysis, the dissertation will propose a conceptual framework and practical recommendations to enhance cloud-base digital forensics practices.

1.3 Methodology

This dissertation will consist of secondary research only. No primary research will be conducted throughout the duration of this dissertation project. The main method of research will be to use ResearchGate, IEEE Xplore, and the UWE Library Search engine to look for reliable academic journal

articles and books related to cloud forensics, cloud forensics challenges, forensic tools, and legal issues. These sources provide insights into challenges, techniques, and regulations in cloud computing. Relevant newspaper articles and industry reports will be used to support the analysis and framework development.

1.4 Dissertation Chapter Structure

This dissertation is made up of four chapters:

- Chapter 1: Introduction
- Chapter 2: Literature Review: Challenges, Techniques, and Legal Implications in Cloud Forensics
- Chapter 3: Conceptual Framework Development
- Chapter 4: Conclusion

Chapter II — Literature Review of Digital Forensics in Cloud Computing – Challenges, Techniques, and Legal Implications

2.1 Challenges faced in Cloud Forensics

The distributed architecture of cloud computing has introduced a completely new forensics challenge compared to traditional forensics **(Montasari and Hill, 2019)**. According to **Montasari and Hill (2019)**, one of the challenges is the jurisdiction problem, and it is illustrated by the Microsoft Ireland case from 2013 to 2018, in which the United States tried to retrieve data from Ireland, leading to a conflict of sovereignty **(Microsoft, no date)**. **Malik et al. (2024)** points out multi-tenancy and the cloud's complexity on how to acquire the log files and difficult to identify the particular data. However, their discussions have not evaluated the scalability of the logging mechanisms and leave the gap of how these challenges evolve along with cloud adoption. These challenges are also corroborated by the National Institute of Standards and Technology (NIST) and stated in **NISTIR 8006 (2014)**, such as determining the location of data and relying on Cloud Service Providers (CSPs) for access to evidence, addressing concerns about jurisdiction and multi-tenancy.

Cinar and Bharadiya(2023) mentioned the reason for legal jurisdiction uncertainty is because cloud service provider (CSP) have separated the data across multiple clouds but they fail to evaluate how the Cloud service provider (CSP) policies accelerate or migrate those issues. **Manral et al. (2019)** have identified the challenges of the 5-step forensics process, which is identification, preservation, collection, analysis and reporting because of the hardware virtualization. Although their framework is detailed, they overlooked the practical implementation barriers, such as cooperation with CSPs. **Alshabibi et al. (2024)** identify the key challenges into three categories: technical, legal and resource-based challenges and propose potential solutions to enhance the effectiveness during the cloud investigation. However, the lack of evidence to support these options has made them difficult to implement.

Vadlamudi et al. (2018) analyze the cloud forensics challenges and the anti-forensics techniques, showing the needs on evidence accessibility, volatility, trust and staff issues. Their insights are valuable but have ignored the systematic issues, such as encryption will obstruct the access. **Egho-Promise et al. (2024)** stated that data privacy and ephemerality are the key issues of cloud environments and the data only be temporary after the incident happens. They also proposed

modifying the existing standards, such as ISO/IEC 27037 and NIST SP 800 series to address the cloud-specific issues, which is ephemeral data and cross-border jurisdiction. Despite their forward-thinking approach, there is an absence of an assessment of the feasibility of global adoption in the context of divergent regulations, such as the General Data Protection Regulation (GDPR)**(REGULATION (EU), 2016)**. **Abuowaida et al. (2025)** indicate the vulnerabilities in the Infrastructure as a Service (IaaS) environment, such as outdated protocols and underlying technology flaws. It also shows deploying machine learning algorithms in the cloud environment would be able to enhance attack detection accuracy.

Ali et al. (2022) address that the cloud service provider (CSP) dependency will be the major challenge to the law enforcement agency, especially when CSPs control the evidence of breaches and might not fully cooperate. Although they proposed a framework to cooperate with Internet Service Providers to reduce the dependency on CSP, they failed to provide any detailed implementation and validation, which could limit its practical utility. **Akter et al. (2020)** highlight some key issues in cloud forensics, such as difficulties in accessing logs that are stored in remote servers and collecting the stable data amid the volatile nature of multi-user cloud environments, proposing blockchain as a solution. Nevertheless, there is a lack of critique of the scalability and cost of the proposal in the study, thus rendering its viability uncertain. **AlSaed et al. (2022)** proposes an integrated framework to enhance cloud forensics investigations by addressing the challenges, such as data collection, integrity and reducing dependency on cloud service provider(CSPs). However, the issues of compatibility across platforms remain unaddressed and it weakens its efficacy. **Al-mugern et al. (2024)** introduces a 6-steps forensics framework: identification, evidence gathering, evidence preservation, evidence analysis, incident documentation, and post-incident investigation to address the unique cloud-specific challenges, such as data decentralization and legal issues. In contrast, the framework has not been tested and it raised doubts about its real-world effectiveness.

These cloud forensics challenges are further reinforced by other points of view. **The Thales Data Threat Report (2024)** shows that 93% of organizations are facing cloud-related threats, highlighting the risk from data dispersion and volatility. The **IBM (2024)** report identified the cost of data breaches averaging around USD 4.88 million and emphasized the urgency of enhancing incident response and detection capabilities. **Dykstra and Sherman (2013)** discussed the technical difficulties in the OpenStack environment, such as the challenge of physical access, yet the FROST tool still remains Cloud Service Provider (CSP) independent. In conclusion, these studies showed the key challenges: jurisdiction **(Montasari and Hill, 2019)**, multi-tenancy **(Malik et al., 2024)**, encryption

(AWS, 2021) and volatility **(Akter et al., 2020)**. However, the frequent absence of empirical depth or practical solutions, such as the limited scalability testing in FROST tools (**Dykstra and Sherman, 2013)**, leaves significant gaps that must be addressed by forensic practitioners.

2.2 Techniques in Cloud Forensics

The unique challenges in cloud forensics, such as data dispersion and the dependence on cloud service provider (CSP), have urged the development of the specialized techniques and tools in cloud forensics for evidence collection and analysis in a dynamic cloud environment (**Montasari and Hill, 2019; Gartner, 2024**). **Manral et al. (2019)** describe the tool "FROST", which is especially designed for the OpenStack environment to capture the virtual machine's data to address the virtualization challenges **(Dykstra and Sherman, 2013).** While "FROST" provides the practical solution, the dependence on Cloud Service Provider (CSP) limits its autonomy and this has not been the subject of adequate scrutiny by the authors **(Dykstra and Sherman, 2013)**. **Malik et al., 2024** advised using Magnet AXIOM Cloud, Cellebrite UFED and AWS CloudTrail for analysing the logs and enhancing the evidence retrieval across the cloud system **(AWS, 2021; Magnet Forensics, 2024)**. **Magnet Forensics (2024)** detailed their cloud capabilities, which can be acquired from different platforms, such as Google, Facebook, iCloud, Dropbox and Microsoft 365 and integrate the data with live sources to streamline the forensics workflow. Moreover, **Malik et al. (2024)** fail to compare the tool's effectiveness between IaaS or SaaS models. In cloud forensics, forensic tools like AXIOM might still have to rely on Cloud Service Provider (CSP) cooperation, which reduces analytical depth and autonomy **(Malik et al., 2024)**.

Alshabibi et al. (2024) introduced machine learning and secure logging to improve the scalability and evidence integrity, which provide a significant potential for large scale investigation in the context of increasing cloud adoption. However, lacking empirical validation, their effectiveness remains speculative. **Abuowaida et al. (2025)** deploy the decision trees and Neural Network in the IaaS environments, showing 90% accuracy in the detection of cyber-attack. However, with the high computational requirements and 30.3% and 38.8% false negative rate raise practicality and reliability issues that have not been fully explored**.** Moreover, **IBM (2024)** reported that the global average cost of the data breach in 2024 was $4.88 million. It has been stated that the potential financial risks associated with undetected attacks due to false negatives, as well as delays from resource-intensive

analyses within forensic tools are the major concern **(Abuowaida et al., 2025)**. **Egho-Promise et al. (2024)** recommend using Artificial Intelligence (AI) driven tools for real-time monitoring to address the data ephemerality issues but lack of integration with existing workflows will limit the practical value**.**

Ali et al. (2022) combined Internet Service Provider (ISP) logs with Information Security and Event Management (SIEM) systems to enhance visibility for law enforcement, reducing the dependency of Cloud service providers (CSPs) in jurisdictional complexity. This method has improved evidence triangulation but those agencies with limited resources may be discouraged from using it due to its complexity and reliance on ISPs **(Ali et al., 2022)**. **Akter et al. (2020)** explored blockchain based logging to handle cloud forensics challenges, for example log access or data volatility management, emphasizing decentralisation can improve the evidence preservation. However, they did not consider the scalability and the cost of using blockchain techniques and raised the concerns about its feasibility. **AlSaed et al. (2022)** used Apache Spark for a large-scale log analysis to reconstruct the incident timelines and reduce the dependency of Cloud Service Provider (CSP), yet the effectiveness is hindered by unresolved compatibility issues with decentralised logs that are not explored in the study.

Al-mugern et al. (2024) proposed a six-step forensics framework: identification, evidence gathering, preservation, analysis, documentation, and post-incident investigation, combined with the log snapshots to address the cloud-specific challenges. This approach can address the decentralized problem but it will raise concern on real-life performance since the framework has not been tested. **Vadlamudi et al. (2018)** suggested using SecLaaS as a solution for client-side to enhance log integrity and address the risk of anti-forensics. However, the scalability of it still has not been tested. **Cinar and Bharadiya (2023)** proposed using Virtual Machine (VM) snapshots to perform forensics analysis but their review does not address specific implementation barriers on this method. The **AWS white paper (2021)** highlighted that CloudTrail is able to monitor encrypted environments under GDPR compliance. However, the key that was managed by the Cloud Service Provider (CSP) had restricted the forensics access. The same challenge also appears in AXIOM's cloud acquisition process**(REGULATION (EU), 2016; Magnet Forensics, 2024, p. 17)**.

2.3 Legal Implications and Regulatory Considerations

Digital forensics are operating in a complex legal environment, which includes three aspects: jurisdictional ambiguity, data privacy regulation and compliance standards. These will challenge the acceptability and integrity of digital forensics **(Montasari and Hill, 2019)**.

Jurisdictional issues are the major challenges of cloud forensics. **Daskal (2018)** highlights the jurisdictional conflicts when the United States authorities tried to access the data that was stored in Ireland, which shows the legal uncertainties in cross-border evidence collection. **Malik et al. (2024)** point out General Data Protection Regulation (GDPR) compliance makes it more complicated during the digital investigation because of the stringent data protection requirements and restrict access to personal data. However, the analysis lacks an in-depth examination of how differences in law enforcement across countries affect forensic practices. **Cinar and Bharadiya (2023)** point out that Cloud Service Provider (CSP) control is the source of legal authority ambiguity, but they did not mention how this affects evidence admissibility in court.

The CLOUD Act was approved in 2018 and brings legal complexity to Europe and the U.K., which allows U.S. law enforcement to force the cloud server provider which is located in the U.S. to provide the data no matter where it is stored **(U.S. Department of Justice, 2019)**. This is achieved by bypassing the more cumbersome Multilateral Convention on the Transfer of Criminal Proceedings between Member States (MLATs) and also enhancing post-Brexit investigation efficiency **(U.S. Department of Justice, 2019)**. However, this benefit does not extend to the broader European Union (EU), where the General Data Protection Regulation (GDPR)'s stringent data transfer restrictions on Article 48, may conflict with the CLOUD Act's extraterritorial reach **(REGULATION (EU), 2016). Daskal (2018)** critiques the CLOUD Act would prioritise U.S. interest first and underscoring its potential to compromise the GDPR by compelling data disclosure from U.S. providers in European Union without enough protections, this tension that was not fully explored by Malik et al. (2024) and Cinar and Bharadiya (2023). After the implementation of the UK General Data Protection Regulation (GDPR) and the CLOUD Act, these bring practical challenges for UK businesses. Especially after the Brexit period, UK businesses have to comply with both GDPR and the CLOUD Act.

However, **Egho-Promise *et al.*, (2024)** have yet to assess the feasibility of these frameworks on a global scale. **Electronic Frontier Foundation (2018)** warned that CLOUD Act lacked international

consensus and it could damage data sovereignty, underlining the need for coordinated legal mechanisms.

Data privacy standards also complicated the forensic process. Data minimisation and "right to be forgotten" under the UK GDPR could erase the crucial evidence, it also conflicts with forensic needs **(REGULATION (EU), 2016)**. **Manral et al. (2019)** highlighted that privacy and the trust issue on Cloud Service Provider (CSP), introduce a Service Level Agreement (SLA) that is aligned with forensics standard. Yet, the proposal still relies on CSP's willingness and there are still remaining practical barriers. **Alshabibi et al. (2024)** emphasise territoriality as a major legal challenge but their solution lacks specificity for harmonising multi-jurisdictional laws, reducing its immediate applicability. **Vadlamudi et al. (2018)** discuss the anti-forensic and the legal gap, pointing out encryption and obfuscation make it more complicated in evidence collection; this limitation has been mentioned by **AWS (2021)** as well. However, they have focused on the technical countermeasure but did not propose a way to reform the current legal framework.

The compliance standards, such as **ISO/IEC 27001:2022** and **ISO/IEC 27037:2012** aim to guide the evidence handling, which limits their applicability to cloud environments. Older standards, such as **NIST SP 800-86 (2006)** and **ACPO (2012)** also lack cloud-specific requirements, reducing their relevance to modern cloud forensics. **(Egho-Promise *et al.*, 2024)** criticise the diversity and cross-border issues of CSPs and propose updates to standards such as ISO/IEC 27037, but do not evaluate the feasibility of global adoption under different legal frameworks **(ISO/IEC 27037:2012)**.

Evidence acceptability is another key issue. **Akter et al. (2020)** points out during court, the jury will not be able to clearly understand the volatile cloud evidence, it probably creates a legal hurdle. Given the complexity of the machine learning model as shown in **Abuowaida et al. (2025)**, if the evidence is delivered by machine learning, it may face judicial acceptance challenges. **Ali et al. (2022)** emphasize that because of the non-cooperation from the Cloud Service Provider (CSP) and the possible delay from the Mutual Legal Assistance Treaty (MLAT), they are proposing the CFF-LEA collaborative approach. However, the framework relies on signature-based detection and it will be limited to identify the novel attack.

In the cloud environment, digital readiness has become a challenge. Especially when Service Level Agreements (SLAs) with Cloud Service Provider (CSP) might not include specific provisions for forensic investigation. **AlSaed et al. (2022)** proposed more robust protocols but they have not

addressed the cross-jurisdictional enforceability. **Al-mugern et al. (2024)** consider the existing legal framework is outdated and require an update but they do not provide a concrete implementation strategy. **The European Union Agency for Cybersecurity (2022)** proposes encryption as a technological barrier against CLOUD Act requests. However, it is not fully addressed in the literature.

2.4 Summary

The literature shows that cloud forensics is facing the serious technical challenges and the legal hurdles, while data dispersion, encryption and jurisdiction complication makes it more complicated when acquiring the evidence. Forensic tools, such as FROST, MAgnet AXIOM Cloud are aimed to solve these issues but still have to rely on Cloud Service Provider (CSP) to cooperate, which limits their scope. The legal frameworks of the GDPR and the CLOUD Act limit data access, which makes it more complex in this area. Many researches lack real-world practical testing and they are mainly focused on the western environment. Future researchers should focus on validating the practical tools and non-western legal perspective to enhance the cloud forensics practices.

Chapter III Conceptual Framework Development

Chapter 3.1 Introduction

Chapter 2 has addressed the key challenges that will affect digital forensic in the cloud environment, including technique, organisational and legal aspects. Major issues would be jurisdictional complexity, such as the Microsoft Ireland case, where U.S. authorities required the data that was stored on an Ireland server, causing a conflict between U.S. legislation and Europe's data protection and leading to the Cloud Act **(Foitzick, 2020)**. Another challenge would be relying on the Cloud Service Provider (CSP), investigators have to rely on CSP to provide the snapshots or the logs. There will be risks of CSP not cooperating or data has been modified during the process **(Alex and Kishore, 2017)**. Multi-tenancy and virtualization makes cloud forensics more complicated because there are multiple users sharing the same physical resources. There will need to be strict isolation to ensure the integrity of the evidence, the nature of virtual machines and shared storage can weaken evidence durability and authenticity **(Malik *et al.*, 2024)**. Meanwhile, log volatility and data ephemerality also raise concern, for example the transient logs might only exist for a certain time or be overwritten quickly, which makes it difficult to preserve the evidence **(Malik *et al.*, 2024)**. Finally, encryption and private measures always obstruct forensics access, even though they aim to protect user privacy, but without the key, investigators could not gain access to the suspect's data. In the forensics aspect, it actually makes evidence collection more difficult **(Alshabibi, Bu dookhi and Hafizur Rahman, 2024)**. All of these challenges show investigators need a comprehensive solution and not just rely on traditional methods.

To address those gaps, this chapter is proposing a conceptual cloud forensic framework with a multi-layers approach, combining technical process, legal and organisation workflow together **(Malik et al., 2024)**. The framework is able to be adapted to different cloud environments, including public, private, hybrid or other cloud services, and also interoperable with different jurisdictions and service providers. The previous frameworks are focused on specific areas, such as technical or legal aspects and this framework is trying to bring it together **(Alshabibi et al., 2024)**. The framework has included clear and workable stages: evidence identification, preservation and presentation, along with the corresponding legal or procedures measures. Moreover, the framework has implemented some new ideas, such as dynamic evidence preservation mechanism, to reduce the effect of volatility data **(Malik et al., 2024)** and embedded the cross-border legal protocol into the investigation process

(Foitzick, 2020). The framework aims to enhance the effectiveness in cloud forensic and ensure it meets the legal requirement.

Chapter 3.2 Framework's structure and components

Phase 1 Incident Identified and scope

First, the process will detect the potential events in the cloud environment and determine the scope. This may be based on the alert from intrusion detection systems or suspicious account activities **(Malik et al., 2024)**. During this phase, investigators have to identify which cloud assets have been involved, such as virtual machines, storage or user accounts, and assess the potential impacts **(Al-mugern et al., 2024)**. The critical part of the forensic scope is to ensure the relative data and the location of the system because it may cross multi-data centres or even different countries. Sooner identify the scope of the incident and be able to guide the future steps, such as knowing the location of the data centre and provide suitable legal strategies **(Foitzick, 2020)**. Log data from Cloud monitoring tools, such as AWS CloudWatch or Azure Monitor can help to identify but these data logs are temporary and require to be quickly secured **(Alshabibi, Bu dookhi and Hafizur Rahman, 2024)**. Moreover, allowing affected Cloud Service Provider (CSP) security teams to get involved when it is appropriate, allows them to prepare for the next step **(Alex and Kishore, 2017)**.

Phase 2 Legal Authorization and Jurisdiction Assessment

Once identified the accident, the next procedure is to ensure investigators have proper legal authorisation to conduct the investigation. Due to the legal complexity in cloud environments, investigators have to ensure which legislation is suitable and have to require a warrant or a court order if necessary **(Egho-Promise et al., 2024)**. For example, if the data are stored in a different country, investigators may need to invoke international cooperation mechanisms, such as Mutual Legal Assistance Treaty (MLAT) or the U.S CLOUD Act **(Foitzick, 2020)**. Any potential conflict between different legal jurisdictions districts will be identified at this stage, for example when transmitting personal data will violate GDPR and develop the strategies to migrate the conflict. The outcome of

this stage is documented all the legal approval, such as search warrant, preservation orders or letters to CSP. With all those documents, investigators can get approval for acquiring the evidence and also be able to demonstrate that all evidence was obtained lawfully **(Malik et al., 2024; Ali et al., 2022)**. Moreover, it is important to cooperate well between legal consultants and investigators; it is essential to allow legal consultants to get involved as soon as possible because this can ensure all actions are under legal authorisations **(Egho-Promise et al., 2024)**. By implementing the legal step into the framework, investigators can be assured that the technical and legal actions are interoperable and all actions can stand up to scrutiny in court because it is conducted under the correct legal framework. Thus, CSP's compliance team should be notified and get involved if necessary because CSP may have policies on how to respond to authority's requests during the investigation process **(Ali et al., 2022)**.

Phase 3 Evidence Collection Orchestration

After defining the scope and obtaining legal permission, the framework will move to evidence collection and introduce an adaptive collection strategy, using different techniques depending on cloud service model and nature of the data.

3.2.3.1 Data extraction with Cloud API

Acquiring evidence by using Cloud Service Provider (CSP) provided API or management console. For Infrastructure as a Service (IaaS) events, this includes taking a snapshot of the virtual machine's disk and memory, such as using AWS's API to create an elastic block storage snapshot or using Azure's tools to acquire the disk image. For Platform as a Service (PaaS) or Software as a Service (SaaS) events, it might require exporting the application log, user activities and database entries **(Dykstra and Sherman, 2013)**. The framework recommended using automatic tools to run these tasks when possible, which can enhance the speed and consistency.

3.2.3.2 Indepence Logs Preservation

Cloud log, such as access log or audit trails of administrative actions are usual volatility. Inorder to reduce the dependency on Cloud Service Provider (CSP), the framework have a independent log preservation approach. For example, when the organisation sets up a cloud service for their business, the log will transfer to an external server or a third-party server in real-time **(Alex and Kishore,**

2017). With this method, if the attacker tries to delete the cloud log or due to the short CSP retention period, there will still be an untempered copy of the log. This way can ensure the best practice, which is to maintain a constant record of system events.

3.2.3.3 Multi-Tenancy-Safe Imaging

In the cloud multi-tenancy environment, when investigators tried to acquire the full physical disk might include other tenancy data, leading to privacy and legal issues. Framework's procedure is to perform a tenancy-scope image, using CSP's capability to export target's virtual machine storage slice and RAM. For instance, using a management capture of a virtual machine's running status, with CSP facilitated internally, can ensure that other user data is not collected **(Malik et al., 2024)**. If a full host snapshot is not avoidable, the framework requires filtered or security masked non-target data with CSP supervision. This addresses the privacy and data isolation challenges in a virtualized environment.

3.2.3.4 Encryption key identified

To address the encryption barriers, it will secure any encryption data or credentials that need for decryption and it will be required to cooperate with Cloud Service Provider's (CSP) key management service (KMS). If a suspicious virtual machine used a cloud management key, investigators can capture its random access memory (RAM) to retrieve the encryption key. With the early planning in the evidence collection phase, the framework can increase the chance of analysing the encryption data and it is no longer inaccessible**(Alshabibi, Bu dookhi and Hafizur Rahman, 2024; Vadlamudi et al., 2018)**. It is important to maintain a chain of custody during the evidence collection process. Framework will be required to use the HASH algorithm, such as SHA-256 for every piece of evidence and detail documented in the evidence's log. Moreover, the framework recommends tamper-resistant storage, for example, writing the evidence to append-only storage or a blockchain record for evidence handling events to enhance the integrity of the evidence **(NISTIR 8006, 2014; ISO/IEC 27037:2012)**.

Phase 4 Evidence Preservation and Storage

After the evidence collection process, it has to be preserved in a secure manner for further analysis. During this stage of the framework, it will focus on protecting the integrity and confidentiality of the evidence and also ensure only authorized investigators can be accessed **(NISTIR 8006, 2014)**. Thus, all evidence artifacts, such as logs, virtual disk images, etc will be transferred to a secure evidence repository **(Egho-Promise et al., 2024)**. In a cloud environment, this will become a specific-cloud secure storage or depends on the sensitivity of the case's local forensics server. The repository should set up strict access control, versioning and immutability features, which only allow the investigation team to gain access and ensure evidence would not be modified or deleted without any detection. In addition, the hash value that is calculated in the collection phrase is used to verify that the evidence remains unchanged to maintain integrity **(ISO/IEC 27037:2012)**. It also recommended duplicating copies to different servers, considering geographic and jurisdiction challenges, having other copies can migrate the risk of data loss or cross-border access issues **(NIST, 2014)**. The framework introduces dynamic evidence retention policies to address cloud-specific challenges, such as data volatility **(Manral et al., 2019)**. If the investigation scope has expanded or discovered new relevant data, the system should be able to dynamically update the retention, allowing the cloud service provider to put a legal hold on the suspect account to prevent data deletion. By proactive preversing the data, the framework is able to minimize the disappearing window of volatility evidence. All the steps are required to be documented to maintain the integrity of the evidence. At this moment, the evidence has been effectively frozen and secured, ready for further examination.

Phase 5 Evidence Examination and Analysis

During this stage, investigators will examine the preserved evidence and find the relevant facts about the incident. The framework supports a wide range of forensic analysis techniques and focuses on the correlated data across sources. For example, by combining the cloud service provider's system record with the login, API call log, and virtual machine's disk image with file system timestamp, and application log to reconstruct event sequences **(Montasari and Hill, 2019)**. According to the different types of data, there will be various tools and techniques for investigators.

Firstly, for virtual machine's disk or memory image, standard digital forensics tools like EnCase or Autopsy, are able to recover deleted documents, analyse artifacts and identify malware. With the Volatility framework, investigators can examine the use of memory and allow them to find the running processes, network connection or the encryption key that is stored in random access

memory (RAM) **(Malik *et al.*, 2024)**. Secondly, for cloud-specific logs and artifacts like AWS CloudTrail log, Azure AD sign-in data or SaaS application audit log, investigators can use the specific script for parsing it. The framework recommends the use of automatic tools to filter massive log data to find any anomalies. Recent research also shows the machine learning trends in the cloud **(Alshabibi, Bu dookhi and Hafizur Rahman, 2024)**. Combined with intelligent analysis, they are able to assist with the massive scanning of cloud logs and flag evidence that is missed by manual review. In addition, if the data are encrypted and investigators retrieve the encryption key, the key will be able to decrypt the content and be ready for examination. If the key is not retrieved, the investigation team might have to try another legal process to address these challenges **(Alshabibi, Bu dookhi and Hafizur Rahman, 2024)**. During the whole analysing process, the framework has documentation of finding, such as the timestamp of data exfiltration or a malicious IP address will be recorded in the analysing logs, and stated the sources. This can ensure forensic transparency and reproducibility, which are important for later court testimony.

One of the most important challenges in the evidence and analysis phase is the cross-sources correlation and cloud contexts. Unlike traditional digital forensics, cloud forensics needs to be involved in linking data from different cloud servers to personal devices **(Montasari and Hill, 2019)**. The framework has suggested investigators combine all the evidence or case database into a central timeline. Investigators are able to find connections by accessing the database since the evidence is coming from different sources and it will only show the full story by combining all of them.

Phase 6 Report Generation and Legal Presentation

After the examination and analysis of evidence, the result of the examination has to become a formal report that is ready for legal scrutiny. The framework is focused on turning the technical analysis into a formal report that can be used by lawyers, judges or other stakeholders. Investigator team needs to prepare a comprehensive forensics report, including the exclusive summary, detailed description of the evidence that was collected and analysed, event timeline and the conclusion of the incident**(Manral et al., 2019)**. Moreover, the report has to clearly state the supporting evidence with the hash value and reference with the evidence identifier inorder to prove the integrity of the evidence**(ISO/IEC 27037:2012)**.

The report ensured the legal acceptability of the evidence; this included verifying the chain of custody of the files, using the method that had generally been accepted in the field **(ISO/IEC 27037:2012)**. If they are using any specified tools or novel techniques, it might require an expert affidavit or a reference to the previous case to prove the validity and to explain to the jury in non-technical terms.The framework also recommends preparing the data presentation in court, such as exhibits with a log's screenshot. In addition, considering cloud incidents usually involve international dimension and different jurisdiction districts. If it is involved multiple countries and investigators might need to prepare separate documentation aligned with each legal system's recruitment and ensure consistency. The framework has made this process smoother since the jurisdictional considerations were addressed in Phase 2 **(Daskal, 2018)**.

Last but not least, part of this phase is to provide recommendations for improving security and forensic readiness. With such forward looking advice, it can create a close loop and improve the future cloud forensics effort. At the end of this stage, the case should be ready for prosecution and ready with a well documented body of evidence and analysis.

Chapter 3.3 Conclusion

These six phases form the main part of the framework and are presented in a linear way for better presentation. However in real life scenarios, they will iterate or overlap during the practical. For example, during the evidence examination and analysis stage (phase 5), investigators might discover new leads and be required to go back to evidence collection orchestration stage (phase 3), or discover some legal issues and be required to go back to phase 2 to get an extra warrant or documentation. The framework allows iteration and is still able to maintain its structure when, every time, there is a need to go back to another phase, and the actions still maintain the best practices and documentation requirements.

Chapter IV Conclusion

This dissertation used secondary research and discussed the complexity of digital forensics in a cloud computing environment. It explored the technical and legal challenges during acquiring, persevering and analysing the evidence in the distributed cloud infrastructures. There are special issues with multi-tenancy, data volatility, encryption, and jurisdictional uncertainty. With a comprehensive literature review and the development of a conceptual framework, this study critically analysed the effectiveness of current forensics tools and methodologies, as well as their effect on current regulatory frameworks, such as the GDPR and the CLOUD Act.

Chapter 4.1 Dissertation finding

4.1.1 What specific challenges limit effective digital forensic investigations in cloud environments?

Cloud environments show the unique challenges in cloud forensics and the major hurdles included the data dispersion across multiple servers, complex encryption for data protection and multi-tenancy. These specific challenges make it difficult when identifying and isolating the evidence, which also means that the evidence is stored in different jurisdictional districts and potentially lead to legal sovereignty conflict, the Microsoft Ireland case exactly shows the same issue. Investigators are also limited in retrieving or accessing the log and data since they require cloud service providers to cooperate with them, which also adds uncertainty and delay while evidence collection. In conclusion, globally dispersed data, encryption complexity, multi-tenancy and jurisdictional ambiguity are the major challenges that impede effective digital forensics investigations in cloud environments.

4.1.2 How effective are the current forensic techniques and tools in addressing these cloud-specific challenges?

Current cloud forensic techniques and tools have provided partial solutions to address these cloud-specific challenges. Specialized tools, such as FROST from OpenStack, which captures the

virtual machine's data to address the virtualisation issue, and Magnet AXIOM Cloud, which acquires evidence from multiple platforms, are able to address the issues of data dispersion and multi-tenancy. Although these tools are able to collect and analyse the cloud evidence, they still have to rely on cloud service providers to cooperate and it is still limiting their scope and autonomy. However, many of these techniques lack real-world validation, and their testing is predominantly limited to Western environments, which might raise concerns about reliability in a diverse and large-scale cloud environment. In conclusion, current tools are able to solve some specific challenges but not completely overcome the dependence on cloud service providers.

4.1.3 To what extent do current legal and regulatory frameworks support or obstruct the process of collecting, preserving, and presenting digital evidence from cloud-based systems?

Existing legal frameworks are obstructing the cloud forensics processes instead of supporting them. They have set the boundary like GDPR on how to handle the evidence, it has a strict limitation on accessing and transferring of digital evidence. When facing cross-border evidence collection, this often slows down the investigation process. Jurisdictional conflict is still a major challenge in the cloud environment. The Microsoft Ireland case shows that there is a lack of international consensus and authorities could not collect the cloud data because of sovereignty and privacy concerns. After the Microsoft Ireland case, the U.S. have announced the CLOUD Act to address the cross-border data request but it also introduces new complexity and is not universally recognized, which result in a mixture of compliance challenges. Investigators are required to carefully handle multiple jurisdiction, service agreements and legal constraints during every step of collecting, preserving and presenting cloud evidence. Overall, the legal framework provided necessary protection on privacy concerns but the fragmentation and restrictives terms always obstructed the efficient cloud forensic process.

4.1.4 What methodologies are currently employed in cloud-based digital forensic investigations and what gaps exist in these practices?

Cloud digital forensics usually follows an adapted standard forensics methodology, but there are still gaps in practicality. Investigators will usually follow the traditional 5-step forensic process, which includes identification, preservation, collection, analysis, and reporting. However, each stage will become more complicated because of the cloud's specific issues, for instant hardware virtualization

and remote data storage. To solve these issues, researchers introduced a framework specifically for the cloud, which included a 6-steps process with detailed incident documentation and post-incident analysis. The framework is unlike the traditional method and it covers the decentralised data and legal issues. However, many of these frameworks remain theoretical or untested, and it is questionable whether they are effective in the real world. In practice, investigation requires cooperation with cloud service providers to retrieve the logs. This lack of a standard cross-platform procedure is causing inconsistent evidence handling, difficulties in applying forensics techniques to a large cloud infrastructure, and uncertainty in maintaining the chain of custody. Literature emphasises that the proposed methods lack depth and validation, which requires the development of a standard forensic process for the cloud environment. In summary, some basic forensic processes are currently employed in cloud-based digital forensics investigations, but there are gaps between the proposed methodologies and their implementation.

4.1.5 What solutions and best practices can be proposed to enhance the efficiency, reliability, and legal compliance of digital forensic investigations in cloud computing?

To enhance the efficiency of cloud forensics, there will need to be a reliable and legal solution that combines integrated technical and procedural aspects. The conceptual framework that develops in chapter 3 aims to enhance the efficiency of the investigation while maintaining the legal standard. This approach adopts the best practices from industries and academic fields, which use cloud-native monitoring and logging tools to ensure the timely capture of evidence, followed by using automatic tools to analyse and address the data dispersion issues. Moreover, the framework also has strict evidence preservation measures to guarantee reliability in court, such as setting up tamper-evident audit trails for digital artifacts. In addition, the researchers recommend a closer collaboration between forensic analysts and cloud service providers to develop clear protocols and service level agreements for evidence collection to reduce dependencies. Furthermore, evidence collection strategies should comply with laws such as GDPR and the ACPO good practice guide to maintain a proper chain of custody. However, the ACPO only provides the foundational principles, and it is not fully integrated with the cloud environment. Last but not least, by implementing these solutions and best practices, investigators are able to enhance the efficiency in handling technical complexities, maintain the evidence integrity and ensure compliance with the multi-jurisdictional requirement.

Chapter 4.2 Limitations and Suggestions for Future Research

The limitation of this dissertation is that it relies on secondary research, and it is suggested that further research on this topic would need to use primary sources to ensure the framework's effectiveness and feasibility, such as expert testimony or practical case analysis. This limited the depth and the practicality of the findings. Moreover, the research mainly uses Western research papers and lacks the legal and technical perspectives of non-Western countries. This would affect the framework's applicability, and future research should focus on international issues, especially those involving different cross-jurisdictional forensics and legal implications. Lastly, since cloud technologies and cyber threats are evolving rapidly, future research must continuously update current forensic methods and tools to ensure their adaptability to emerging challenges.

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Appendix A Critical Appraisal

Appendix A1 - Topic Choice

Cloud computing has become the most important component of modern IT architectures. Individuals rely on the cloud service for the convenience and businesses keep adopting their services into the cloud for their flexibility and efficiency. This shift changed the way we store data, cooperate and daily operation. Under this situation, I was thinking: is digital forensics keeping pace with this trend? This inspired me to choose cloud forensics as the dissertation topic. When I started to explore this area, I found out there are less studies than I expected. Technology is evolving everyday but the current forensics framework and studies seem insufficient and outdated. These gaps inspired me to combine cloud computing and digital forensics for my dissertation topic.

Throughout the three years of digital forensics learning, we have to follow the key principle: maintain chain of custody, detailed documentation meticulously for court. These guidelines ensure our work can hold up under scrutiny. However, when I started doing some research on cloud forensics, I realised there do not have any official framework for cloud specific. On the other hand, investigators have adopted the traditional framework and tweaked them to fit the cloud specific environment. The finding led me to choose this topic as my research and that is what I hope it contributes to.

Appendix A2 - Reflection on Project Management

I knew this project would be the biggest task and it would take most of the time of my final year. I knew time management would be the largest challenge that I faced and I've set up a timeline that I hope I can follow during the period. I have set up a gantt chart (see Appendix B5) and break it down into manageable stages. However, I underestimated the workload from other modules and caused me to fall behind the schedule. Furthermore, working during the holiday makes my progress even slower than I expected. Although I have requested to have an extra day off due to the holiday, I still have to work during the holiday period, which makes my planning delayed further. Even though I still manage to squeeze in writing the draft and reading with the time I have left, a gap already opened

up between my timeline and where I actually was. It became clear that I need a new strategy to work on the dissertation and my other module's assessment.

In the second semester, I started to decide to work little by little and I realised that working a little each day, rather than tackling huge chunks all at once. It might suit me better, especially with multiple assessments piling up. I set daily goals and aim to finish one small chapter per day. After I follow this pattern I start keeping up on my timeline and making steady progress everyday.

Early on, I didn't request any supervisor meeting for the first few months because I was just keeping reading and taking notes and it didn't feel like I had anything substantial to show. It wasn't until January that I made any real progress, and even then I waited until March when I had a couple of chapters finished before I reached out. However, I think that's actually a terrible idea because a supervisor can provide recommendations on what I'm facing and having those check-ins could push me to hit smaller milestones along the way.

Appendix A3 – Dissertation Strengths

I believe the key strength in this dissertation is the conceptual framework. This framework is highly adaptable in different situations and it allows investigators to revisit different phrases according to their needs, which provide flexibility to fit real-world applications. It also addressed the current cloud-specific challenges and designed to incorporate future updates, offering a practical solution for investigators working in the dynamic cloud environment.

Another strength would be the interdisciplinary approach in this dissertation. In my literature review, I noticed there are not many studies that include different areas altogether. Most of them are only focused on one or two sectors, such as challenges, technical aspects or legal implications. By integrating these aspects, it will easily have a more comprehensive perspective. In addition, while comparing two major regulation frameworks, GDPR and the CLOUD Act, it will allow the framework to address the most critical challenges in cloud forensics which can enhance the dissertation relevance to both forensics practitioners and policymakers.

Appendix A4 – Dissertation Weaknesses

A weakness of this dissertation was it completely relied on secondary research, which limits the depth of insight of real-world application. For example, interviews with forensics experts, cloud service providers or legal professionals, these are the findings that could not be explored by the second research. Without the primary data, it might show my findings are less applicable to the real-life scenario.

A second weakness is that the dissertations mainly focused on western's legal and technical environments, which might limit the applicability to the global. Legal systems and cloud adoption might be different in Asia or Africa countries, which could reduce the framework's relevance in the global environment.

Finally, since the study is using secondary research, it would lead to the framework lack of validation. The framework is based on the literature and it has not been tested, practical experiments or expert review. The absence of the validation would raise concerns about the framework feasibility and effectiveness in real-world investigation.

Appendix B – Supporting Documents

Appendix B1 – Project Proposal

Project Proposal

Applicant Details	
Name of Student:	CHUN-YIN NG
Student Number:	22034405
Student Email Address:	Chun7.Ng@live.uwe.ac.uk
Date	30/09/2024

Project Details	
Proposed Project Title:	Digital Forensics in Cloud Computing: Challenges, Techniques, and Legal Implications.

Project Type (Tick correct project type)			
Dissertation Project:	☒	**Technical Project:**	☐

Aims and Objectives
Describe the key aims of the project and list the key objectives

Key aim:

- Analyse the challenges in conducting Digital Forensics in a Cloud environment.
- Explore and evaluate current techniques and tools for Cloud-based Forensics.
- To investigate the legal and regulatory implications of cloud forensics.
- Propose solutions and best practices for enhancing cloud-based digital forensics.
- Investigate the methodologies employed.

Key Objectives:

- To Identify and Categorize Cloud-Specific Challenges in Digital Forensics.
- To Examine and Evaluate Existing Forensic Tools and Techniques for Cloud Environments.
- To Assess Legal and Regulatory Frameworks Impacting Cloud Forensics.
- To Develop a Theoretical Framework or Set of Guidelines for Cloud-Based Forensic Investigations.

Outline of Project
Describe briefly what you intend to investigate, and for technical projects, the system you intend to produce. Please identify any specialized equipment you are likely to need, and how access to this will be achieved.

What intend to investigate:

The project aims to investigate the approaches to Digital Forensics in Cloud computing. This includes exploring the unique challenges, techniques and legal implications each region faces when conducting cloud-based forensic investigations. Special emphasis on comparing the legal framework, forensics tools, and methodologies used in both regions, and how these impact the ability to extract, preserve, and analyze digital evidence in the cloud.

Specialized equipment:
In conducting cloud forensics investigators use different tools and platforms based on local practices, legal requirements, and available resources.

Forensics Tools:

- EnCase cloud forensics
- Volatility
- AWS
- Cellebrite Cloud Analyzer

Key References and Literature Review
Please identify at least two key references in UWE Harvard format that are informing the direction for your project. Perform a mini-literature review for these references detailing their significance.

Key references:

1. Ramakrishnan, G. and Haqanee, M. (2024) Cloud Forensics Demystified : Decoding Cloud Investigation Complexities For Digital Forensic Professionals. 1st ed. Birmingham: Packt Publishing.

The book serves as a comprehensive view for digital forensic professionals seeking to comprehend the intricacies of cloud-based investigations. As the adoption of cloud services continues to grow, this book addresses the requirement for specialized expertise in cloud forensics. It encompasses both conventional forensic methodologies and contemporary cloud-native tools. The book emphasizes practical approaches for investigating incidents across different cloud platforms, including Amazon Web Services (AWS), Azure, Google Cloud Platform (GCP), Microsoft 365, and Google Workspace.

2. Quick, D., Martini, B. and Choo, K.K.R. (2014) *Cloud storage forensics*. 1st edn. Edited by B. Shavers. Waltham, MA: Syngress.

The book that delves into the intricate challenges and methodologies for conducting forensic investigations within cloud storage environments. This comprehensive resource elucidates the transformative nature of digital forensics in the context of data migration to cloud storage. It elucidates the tools, techniques, and the unique legal and technical considerations that emerge during cloud-based investigations.

Reading list:

3. Martini and Choo (2012) An Integrated Conceptual Digital Forensic Framework For Cloud Computing. Digital Investigate 9 (2), pp. 71-80. [Accessed 28 October 2024].

4. Casey (2012) Cloud Computing and Digital Forensics. Digital Investigate 9 (2), pp. 69-70. [Accessed 28 October 2024].

5. Malik et al. (2024) Cloud Digital Forensics: Beyond Tools, Techniques, and Challenges. .24 (2), p. 433. [Accessed 28 October 2024].

6. Choo et al. (2016) *Cloud Forensics: State-of-the-art and Future Directions.* Digital Investigate [online]. 18, pp. 77-78. [Accessed 28 October 2024].

7. Pichan et al. (2015) *Cloud Forensics: Technical Challenges, Solutions and Comparative Analysis.* Digital Investigate [online]. 13, pp. 38-57. [Accessed 28 October 2024].

8. Ho et al. (2018) Following the Breadcrumbs: Timestamp Pattern Identification For Cloud Forensics. *Digital Investigate* [online]. 24, pp. 79-94. [Accessed 28 October 2024].

9. Choo Et, A.L. (2024) Grand Theft Api: A Forensic Analysis of Vehicle Cloud Data. *Forensic Science International: Digital Investigation* [online]. 24, p. 301691. [Accessed 28 October 2024].

CyBOK Justification
Please explain with reference to the CyBOK why you consider your project falls within particular CyBOK domain(s). You should refer to the Cyber Security Body of Knowledge (CyBOK) Knowledge Areas, which are available to view online at https://www.cybok.org/knowledgebase/
CyBok domains: - Digital Forensics - Cloud Security - Law & Regulation - Human Factors - Cyber-Physical Systems Security

Appendix B2 – Supervisor Meeting Notes

UFCE8A-30-3: Project and Dissertation

Meeting Notes

Title of Project:	Digital Forensics in Cloud Computing: Challenges, Techniques, and Legal Implications.
Student name:	CHUN-YIN NG
Supervisor name:	Jay Murphy

Date	Meeting Notes	Actions
25 November 2024	First meeting with supervisor Jay.	Highlight the time and actions that we have to do and things expecting to be done before the end of January.
16 December 2024	Second meeting with supervisor Jay on Team meeting	Discussed things that should be done during the Christmas holiday and provided recommendations on how to find good quality research papers and how to use additional tools to save the research and references it.
9 March 2025	Third meeting with Jay	First look at the dissertation and provide feedback on adjusting the order of aim and objectives and future changes based on the current progress. Thus also discuss how to manage the time since the deadline is coming.
20 March 2025	Forth meeting with Jay	Second look on the dissertation and progress was a bit slow but able to catch up. Remember to submit the final draft before the end of the supervisor period.

You are free to follow your own way of keeping notes. Ideally you should keep the date of the meetings, what was discussed, and the way you addressed, or will address it.

Appendix B3 – Progress Review December

December Progress Review

Project Details	
Student Name:	CHUNYIN NG
Student Number:	22034405
Student Email Address:	Chun7.Ng@live.uwe.ac.uk
Date	11 December 2024
Supervisor Name	Jay Murphy
Project Title:	Digital Forensics in Cloud Computing: Challenges, Techniques, and Legal Implications

Project Overview
· *A brief description of the project aims and objectives (refined from the proposal)*
Refine the project aims and objectives from the proposal. Four aims remain unchanged with analysis of the specific challenges, techniques and tools, legal and regulatory implications and best practices in the cloud environment. However, a theoretical framework was added and will be proposed after the literature review to enhance the forensics practice in cloud environments.

Progress To Date
· *Summary of work complete so far:* o ***Dissertation -*** *Literature review: scope, quality and initial findings. Development of the theoretical framework or research questions* o ***Technical –*** *Literature review, summary of requirements analysis, initial designs* · *Key milestones achieved* · *Challenges Encountered and how they were addressed* · *Ethical Approval Status*

Summary of work complete so far:

Dissertation:

- Literature review still in progress; mainly focus in cloud forensics area and looking for any specific framework about cloud forensics.
- Scope included cloud specific challenges, tools, legal implications.
- Framework in the cloud forensics area is rare and usually organisations will use a traditional framework and twist it into a cloud forensics framework.

Key milestones achieved:

- Outline of the project was done
- Introduction, Aim and objective completed

Challenges faced:

- Research on cloud forensics framework
- How to select suitable research for the project
- Time management

Ethical approval status:

- Passed

Feedback and Reflection
· *Feedback received from the supervisor or peers* · *Changes or adjustments made to the project based on feedback*
Feedback received from the supervisor or peers: - Only one group meeting with the supervisor in December and mainly to discuss the work that needs to be done after the holiday. - Provide recommendations on using what tools and where you can find high quality research paper.

Plans for the Next Phase
· *Planned activities for the next 6–8 weeks* · *Anticipated challenges and potential solutions*
Planning: - Reading on all relative research paper - create second to third draft on the literature review section Challenges and potential solution: - Finding good quality papers and up to date research - Relative research papers - Using Research Rabbit or similar tools to find relative papers

Appendix B4 – Progress Review February

February Progress Review

Project Details	
Student Name:	**CHUNYIN NG**
Student Number:	**22034405**
Student Email Address:	**Chun7.Ng@live.uwe.ac.uk**
Date	**17 January 2025**
Supervisor Name	**Jay Murphy**

Project Title:	Digital Forensics in Cloud Computing: Challenges, Techniques, and Legal Implications

Progress Update
• *Provide a concise summary of the progress made since the December Review* o *Dissertation – Findings, data collection (if applicable), and chapter drafts* o *Technical – Literature review, Prototyping, testing and implementation progress*
Findings: Some frameworks that applied to cloud forensics, legal legislation related to the cloud environment, standards or gaps that appear in the cloud forensics environment. Specific challenges are not entirely solved and still leave some gaps. **Chapter drafted: Chapter 2, Challenges, techniques.** **Working on Chapter 2, Legal implication**

Key Milestones Achieved
• *List of completed sections, modules, or objectives met*
- Completed sections: Introductions, aim and objective **- Objectives met: Finish reading on the literature and able to move forwards to chapter 2, legal implication.**

Challenges and Problem Solving
• *Major Challenges Encountered Since December – Technical, theoretical, or logistical difficulties* • *How were these addressed? – Solutions, adaptations and changes in approach*

Major Challenges encountered since December:

- Find up-to-date sources and related papers

- How to be referencing correct

- Time management

Solutions:

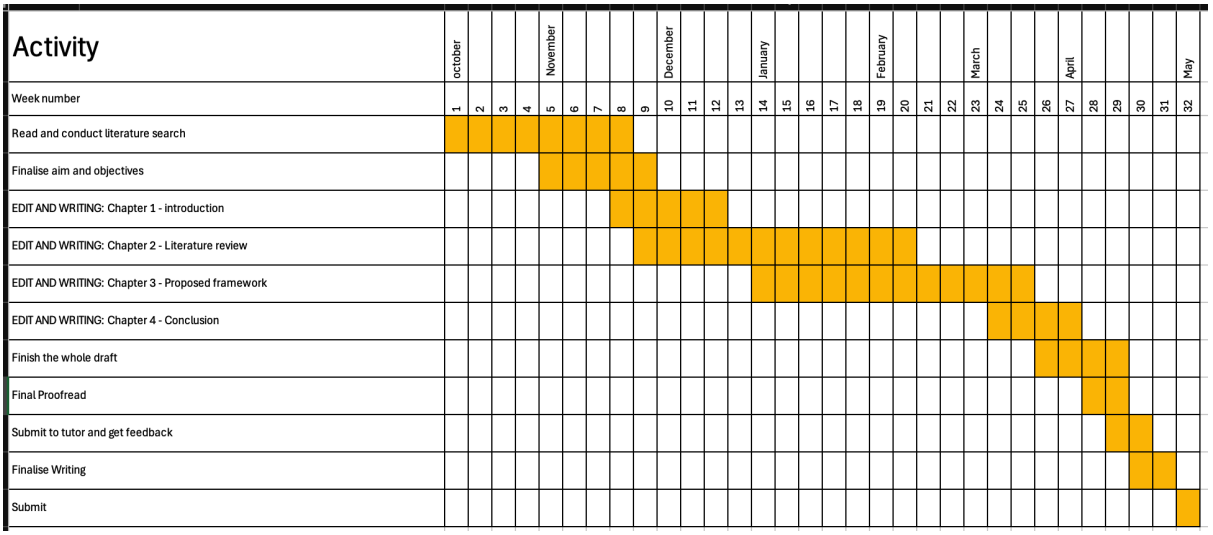
- Use Research Rabbit and Zotero for finding great sources.

- Zotero can generate UWE references but might be not accurate; Google extensions are more accurate to the UWE reference guide; UWE reference guide can double check the reference style when there are some references that cannot be generated by the google extension.

- Setting an alarm and setting a small goal for myself to achieve every day. Build up the content little by little.

Remaining Work and Completion Plan
• ***Outstanding Tasks – What remains to be completed*** • ***Estimated Timeline for Remaining Work – Breakdown of tasks with estimated completion dates*** • ***Current Risks to Completion – Any delays or obstacles foreseen***
Outstanding Tasks **- Chapter 2 Legal implementation** **- Chapter 3 Framework** **- Chapter 4 Conclusion** **Estimated Timeline for remaining work** **- Chapter 2 (Estimated finish on 3 March 2025)** **- Chapter 3 (Estimated finish on 24 March 2025)** **- Chapter 4 (Estimated finish on 3 April 2025)** **Current risks to completion** **- Not really having any risks at this stage.**

Student Reflection

• ***Confidence in Completion – Do you feel on track for submission*** • ***Support or Additional Resources Needed – Any assistance required to help you complete the project?***
Confidence in completion: **-Feel a bit to fall behind the plan and worry about the framework development.** **Support:** **- Not really needing assistance.** **- Time is required.**

Appendix B5 – Gantt Chart